

# Task 7

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In recent years, **Distributed Denial of Service (DDoS)** attacks have evolved from simple bandwidth floods into highly sophisticated, multi-vector cyber weapons capable of crippling even the most resilient infrastructures. These attacks exploit massive botnets, compromised IoT devices, and reflection-based amplification to overwhelm targets' resources and force critical services offline. Unlike traditional cyber intrusions that aim to steal data, DDoS attacks focus on **availability disruption** targeting the reliability and reputation of organizations.

### **1. Aisuru IoT Botnet Attacks (2024 – 2025)**

#### **Target:**

Large ISPs, gaming networks, and several infrastructure providers across the U.S. and Europe. Some media and security research websites were also temporarily affected.

#### **Technology Used:**

Based on **Mirai-style IoT malware**, Aisuru hijacked hundreds of thousands of unsecured routers and smart devices. It launched **volumetric L3/L4 floods** using UDP, DNS amplification, and raw packet floods, reaching bursts exceeding **10–30 Tbps**.

#### **Attacker's Motive:**

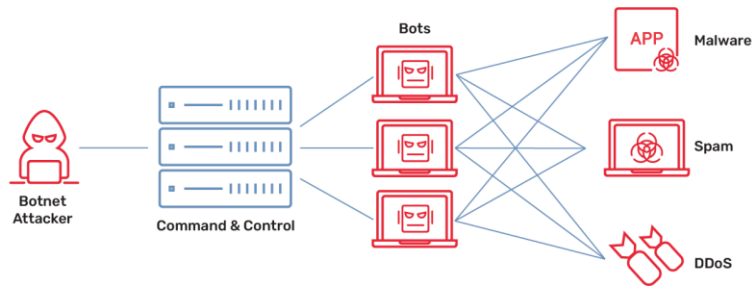
Likely financially motivated botnet operators testing capacity and renting out DDoS as a Service access. Also an attempt to establish dominance in underground forums.

#### **Overall Impact:**

Temporary slowdowns and outages for ISPs and gaming platforms; increased latency for CDN providers during mitigation phases; massive operational cost spikes.

#### **Defensive Strategies:**

- Upstream scrubbing and multi-CDN anycast distribution
- ISP-level ingress filtering (BCP 38)
- Automated rate limiting and burst detection
- Segmented architectures and resilience drills
- Stronger IoT device regulation and firmware patch enforcement



## 2. Cloudflare-Mitigated Megafloods (2025)

### Target:

Cloudflare's enterprise customers including e-commerce sites and online gaming services faced short but extreme DDoS spikes.

### Technology Used:

Distributed botnets launched **multi-vector floods** combining **SYN, UDP, and DNS reflection**. The largest recorded hit over **11.5 Tbps**. Attacks lasted only seconds but were highly optimized to exploit burst-based capacity limits.

### Attacker's Motive:

Likely stress-testing CDN defenses, probing mitigation limits, and conducting extortion campaigns ("Pay or stay offline").

### Overall Impact:

Minimal service downtime due to Cloudflare's automated scrubbing centers, but significant increase in global backbone traffic; proved that multi-Tbps attacks are becoming "normal."

### Defensive Strategies:

- Real-time traffic analytics and adaptive filtering
- Multi-region load balancing to absorb short bursts
- Enhanced reflection/amplification abuse prevention
- Dynamic firewall rules synced with attack intelligence feeds

## 3. Gcore / Gaming-Host 6 Tbps Incident (2024)

### Target:

A European gaming and hosting provider (Gcore's customer base) experienced one of the largest region-recorded DDoS events.

### Technology Used:

**Botnets of compromised consumer devices** launched **UDP flood attacks**, including game protocol-specific floods and reflection vectors (NTP/CLDAP).

**Attacker's Motive:**

Likely revenge or competition suppression in the gaming scene; possibly a criminal service hired to disrupt major game launches.

**Overall Impact:**

Partial downtime for several hosted gaming platforms; increased bandwidth cost; triggered large-scale peering reroutes across European IXPs.

**Defensive Strategies:**

- Dedicated scrubbing nodes near gaming PoPs
- Geographically distributed firewalls to filter at edge
- Source-rate limiting and anomaly-based detection
- Strengthened SLAs for DDoS-resilient hosting

## 4. Microsoft Azure / Microsoft 365 Outage (2024)

**Target:**

Microsoft's public cloud infrastructure particularly the Azure portal, Outlook, and Microsoft 365 Online services.

**Technology Used:**

A **coordinated L7 DDoS attack** combining HTTP request floods and traffic pattern manipulation. Attackers exploited unoptimized edge filters during a global load-balancer reconfiguration.

**Attacker's Motive:**

Attributed to "Anonymous Sudan" hacktivist group claiming retaliation for geopolitical reasons and U.S. policy positions.

**Overall Impact:**

Users across multiple continents faced multi-hour service disruption and degraded authentication. Microsoft admitted that over-aggressive mitigation caused collateral blocking.

**Defensive Strategies:**

- Layer-7 traffic profiling and behavioral baselines
- Multi-vendor scrubbing with fallback routing policies
- Better failover testing to avoid self-inflicted downtime
- Real-time coordination between SOC and load-balancing teams

## 5. Hacktivist Campaigns (KillNet / Anonymous Sudan – 2023 to 2025)

**Target:**

Government portals, media outlets, and airports in NATO, Ukraine-supporting, and Middle-Eastern nations.

**Technology Used:**

Botnets and crowdsourced “voluntary” attack tools (e.g., *DDoSia*), sending **HTTP floods and DNS queries** coordinated via Telegram channels. Attacks are sustained by propaganda drives rather than pure network power.

**Attacker’s Motive:**

Political hacktivism to protest Western policy, disrupt critical communication channels, and amplify ideological messaging.

**Overall Impact:**

Temporary takedowns of government websites, minor airport and payment disruptions, and heavy media coverage amplifying psychological impact.

**Defensive Strategies:**

- National-level DDoS coordination centers (like CERTs)
- Geo-fencing of malicious IP ranges during campaigns
- Fast CDN cache recovery and rate-limiting at WAF level
- Cyber-hygiene awareness to prevent panic and misinformation